6.6.4 Mutation

Although mutation is the original source of all genetic variation, mutation rate for most organisms is low. Hence new mutations on an allele frequencies from one generation to the next is usually not large.

6.7 Hardy - Weinberg Principle

In nature, populations are usually evolving such as the grass in an open meadow, wolves in a forest and bacteria in a person's body are all natural populations. All of these populations are likely to be evolving some of their genes. Evolution does not mean that the population is moving towards perfection rather the population is changing its genetic makeup over generations. For example in a wolf population, there may be a shift in the frequency of a gene variant for black fur than grey fur. Sometimes, this type of change is due to natural selection or due to migration or due to random events.

First we will see the set of conditions required for a population not to evolve. Hardy of UK and Weinberg of Germany stated that **the allele frequencies in a population are stable and are constant from generation to generation in the absence of gene flow, genetic drift, mutation, recombination and natural selection.** If a population is in a state of Hardy Weinberg equilibrium, the frequencies of alleles and genotypes or sets of alleles in that population will remain same over generations. Evolution is a change in the allele frequencies in a population over time. Hence population in Hardy Weinberg is not evolving.

Suppose we have a large population of beetles, (infinitely large) and appear in two colours dark grey (black) and light grey, and their colour is determined by 'A' gene. 'AA' and 'Aa' beetles are dark grey and 'aa' beetles are light grey. In a population let's say that 'A' allele has frequency (p) of 0.3 and 'a' allele has a frequency (q) of 0.7. Then p+q=1.

If a population is in Hardy Weinberg equilibrium, the genotype frequency can be estimated by Hardy Weinberg equation. $(p + q)^2 = p^2 + 2pq + q^2$

p^2 = frequency of AA

2pq= frequency of Aa

q^2= frequency of aa

p = 0.3, q = 0.7 then,

$p^2 = (0.3)^2 = 0.09 = 9$ % AA

$2pq = 2(0.3)\ (0.7) = 0.42 = 42$ % Aa

$q^2 = (0.7)^2$ 0.49 = 49 % aa

Hence the beetle population appears to be in Hardy- Weinberg equilibrium. When the beetles in Hardy- Weinberg equilibrium reproduce, the allele and genotype frequency in the next generation would be: Let's assume that the frequency of 'A' and 'a' allele in the pool of gametes that make the next generation would be the same, then there would be no variation in the progeny. The genotype frequencies of the parent appears in the next generation. (i.e. 9% AA, 42% Aa and 49% aa).

If we assume that the beetles mate randomly (selection of male gamete and female gamete in the pool of gametes), the probability of getting the offspring genotype depends on the genotype of the combining parental gametes.

Hardy Weinberg's assumptions include

No mutation – No new alleles are generated by mutation nor the genes get duplicated or deleted.

Random mating – Every organism gets a chance to mate and the mating is random with each other with no preferences for a particular genotype.

No gene flow - Neither individuals nor their gametes enter (immigration) or exit (emigration) the population.

Very large population size - The population should be infinite in size.

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